PulseIQ HR Analytics

Phase 2 — Data Analytics Project Report

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Track	Data Analytics
Repository	github.com/ShivamSingh1208/Data-Analytics_project
Git branch	main
Git tag	phase-2 — pending creation after final validation
Reporting period	April 2025 – September 2026 for budget and KPI source data

Executive summary

Phase 2 extends the PulseIQ HR Analytics warehouse with attendance and leave metrics, attrition metrics and grouped risk summaries, productivity metrics, payroll reconciliation, and department payroll budget analysis. The work uses PostgreSQL tables and SQL views, Python loaders, source-data checks, and manual comparisons. Seven existing automated ETL tests passed. The report records verified results and flags remaining reconciliation and KPI-target validation work rather than presenting it as complete.

Task status and evidence

Workstream	Status	Evidence / remaining work
Attendance & leave	Implemented; verified	SQL views created. Monthly attendance, department metrics, leave status totals and date checks reviewed.
Attrition	Implemented; verified	891 monthly department records and 430 risk predictions loaded. Group risk reporting suppresses groups under 10.
Productivity	Implemented; verified	16 teams, 38 sprints and 28,868 tasks loaded. Cancelled work excluded from key completion metrics.
Payroll reconciliation	Partial	18 months checked. May 2025 and Feb 2026 differ from official reference; discrepancies documented.
Budget & KPI targets	Partial	198 budget rows and 180 KPI target rows loaded. Three budget source checks passed; complete target comparisons remain.

Verified Results

Attendance, attrition, productivity and payroll

Attendance and leave

Monthly attendance ranged from 92.89% in May 2026 to 95.51% in July 2025. Finance & Accounts recorded 90.82% across 6,076 scheduled days, and HR recorded 92.51%. These are investigation signals, not proof of a cause. The leave warehouse contains 3,439 records; leave-status totals were checked. Seven approved medical-leave records each span 130 days (910 days in total) and should be validated with the source owner before interpretation.

Attrition

The headcount source contains 891 department-month records from January 2020 through September 2026. The risk source contains 430 predictions, including 43 High and 86 Medium predictions in the September 2026 snapshot. Risk predictions are not actual exits and are not presented as certainty. The March 2026 snapshot had only five predictions and is suppressed from group reporting under the minimum-group-size rule.

Productivity

The task warehouse contains 28,868 tasks across 16 teams and 38 sprints. There are 24,139 Done tasks, 898 Cancelled tasks and 999 tasks with Blocked status. Overall on-time completion was 92.86% among completed tasks with a valid on-time result; average valid cycle time was 4.80 days. QA Automation had a 94.01% on-time rate, while Growth Marketing had 91.30%. These differences do not establish why team performance varies.

Payroll reconciliation

The official reference covers 18 months. Sixteen months match the monthly employer-cost and employee-count reference. May 2025 is short by one employee and ■61,813.15; February 2026 is short by one employee and ■154,811.89. Both differences match duplicate employee/run pairs rejected by ETL. Both records in each pair were rejected, so the cleaned warehouse is short by one employee in each month. No rejected record was restored without approval.

Budget analysis

The 198 department-month budget rows total ■862,030,000.00. Cleaned payroll employer cost totals ■861,369,767.38, giving a net variance of –■660,232.62 (actual minus budget). Largest net overspends are Data & AI ■992,768.20 (0.90%), Customer Support ■482,050.62 (1.01%), and Product & Design ■453,539.93 (0.65%). Engineering had the largest net underspend at ■2,054,508.87. Three department-month budget values matched the original CSV.

Quality, Lessons and Next Steps

Testing and quality controls

Automated tests: 7 passed in 0.78 seconds using the existing ETL test suite. Tests cover reversed leave dates, date-key consistency, duplicate identifiers, negative payroll amounts and open-ended validity dates.

Warehouse checks: Source-to-warehouse department/month counts and three manual budget checks were reviewed. The payroll budget view contains 198 rows across 11 departments and 18 months. KPI targets contain 180 rows across 10 departments and 18 months.

KPI target caveats: 18 task-completion targets are NULL and must remain missing. Attendance targets are stored as 0.94 and should be converted to 94% for comparison with percentage-point attendance metrics. The units for overtime targets must be confirmed before comparison. Actual-versus-target performance comparisons are not yet complete.

Problems and resolutions

- Source budget values included comma separators; the loader removes commas before numeric conversion.
- A pre-existing SQL view had a different column layout; it was safely recreated without deleting payroll or budget data.
- Payroll duplicate pairs caused two monthly reconciliation gaps; the records remain excluded pending data-owner confirmation.
- Missing KPI targets are preserved as NULL instead of being fabricated.

Learning reflection

This phase strengthened my practical understanding of loading source data into PostgreSQL, designing SQL metric views, validating keys and totals, testing ETL rules, and reconciling payroll with an official reference. I also learned to distinguish actual results from model predictions, preserve missing values, and document unresolved issues instead of hiding them.

Next steps

1. Confirm the authoritative payroll records with the data owner and resolve the two monthly reconciliation differences through an approved process.
2. Confirm KPI target units and build actual-versus-target views for attendance, overtime, attrition and task completion; keep missing task targets unavailable.
3. Capture fresh screenshots of Phase 2 analyses and attach evidence to the repository.
4. Re-run automated and warehouse checks, review Git status, commit the Phase 2 files, create the phase-2 tag, and push the branch and tag to GitHub.

Submission note: This report reflects verified work as of 27 October 2026. The Phase 2 Git tag has not yet been created, and the final KPI target comparisons and fresh Phase 2 screenshots remain outstanding.